WORKED SOLUTION

STEP III 1994, Question 10

Mechanics • Relative Velocity

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Read the question before you go on. It is in the STEP Support Programme's question database:

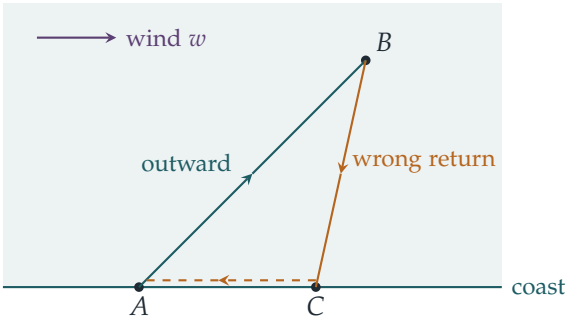
step.maths.org/questions/94-s3-q10

The key idea

Everything is vector addition: the plane's velocity over the ground is its air velocity plus the wind velocity. Put the coast along the x-axis with Alphaport at the origin, and write the outward air velocity in components. Reversing the heading reverses the air velocity but *not* the wind, so the northward and eastward parts of the motion behave very differently on the way back, and that is the whole story of the first part. For the second part, the two correct ground speeds come from the same quadratic, which lets Vieta's formulas do the work.

Solution

We model the plane as a particle moving with constant air velocity of magnitude v, relative to air moving with constant velocity w due east (a wind *from* the West blows *towards* the East). Take Alphaport A as the origin, x east, y north, so the coast is the line $y = 0$ and the island lies in $y > 0$. Betaville B is at $(\frac{d}{\sqrt{2}}, \frac{d}{\sqrt{2}})$. Speeds are in $\mathrm{km\,h^{-1}}$ and times in hours. (The paper calls the places Alphabet/Alphaport and Betaville/Betaport; these are the same places.)



The outward flight

Let the outward air velocity be (a, b), so $a^2 + b^2 = v^2$. The ground velocity is $(a + w,\ b)$, and it must point north-east, so its two components are equal and positive:

$$a + w = b > 0.$$

She covers the y-distance $\frac{d}{\sqrt{2}}$ at y-speed b, and the outward time is t, so

$$bt = \frac{d}{\sqrt{2}} = (a + w)t. \tag{1}$$

The wrong return

With the opposite heading her air velocity is $(-a, -b)$, but the wind still adds $(w, 0)$, so her ground velocity is

$$(w - a,\ -b).$$

Ask yourself Where does she reach the coast, and on which side of Alphaport?

The northward motion is unaffected by the wind: her y-coordinate decreases at rate b from $y = \frac{d}{\sqrt{2}}$, so by (1) she reaches $y = 0$ after exactly time t, the same as the outward flight. In that time her x-coordinate changes by $(w - a)t$, so she crosses the coast at

$$x = \frac{d}{\sqrt{2}} + (w - a)t = (a + w)t + (w - a)t = 2wt,$$

using (1) again. So she lands at C, a distance $2wt$ *east* of Alphaport: going out, the wind pushed her east and her heading was chosen to cancel that, but coming back the heading's correction is reversed while the wind's push is not, so the drift doubles. Since $2wt > 0$, turning West along the coast is indeed the right way.

Flying due west into the wind at airspeed v, her ground speed is $v - w > 0$ (as $v > w$). The coastal leg takes $\dfrac{2wt}{v - w}$, so the whole return takes

$$t + \frac{2wt}{v - w} = \frac{(v - w) + 2w}{v - w}\, t = \left(\frac{v + w}{v - w}\right) t \text{ hours},$$

as required.

The correct return

Write a ground velocity along the line AB as $\pm(p, p)$, where $\sqrt{2}\,p$ is the ground speed. The air velocity is the ground velocity minus the wind, and has magnitude v.

Outward, ground velocity (p_1, p_1), air velocity $(p_1 - w,\ p_1)$:

$$(p_1 - w)^2 + p_1^2 = v^2 \iff 2p_1^2 - 2wp_1 + (w^2 - v^2) = 0.$$

Back, ground velocity $(-p_2, -p_2)$ with $p_2 > 0$, air velocity $(-p_2 - w, -p_2)$:

$$(p_2 + w)^2 + p_2^2 = v^2 \iff 2(-p_2)^2 - 2w(-p_2) + (w^2 - v^2) = 0.$$

So p_1 and $-p_2$ are both roots of the same quadratic $2p^2 - 2wp + (w^2 - v^2) = 0$. Its roots have product $\frac{w^2 - v^2}{2} < 0$, so one is positive and one negative; p_1 is the positive one and $-p_2$ the negative one, so they are the two distinct roots. By Vieta's formulas,

$$p_1 + (-p_2) = w, \qquad p_1(-p_2) = \frac{w^2 - v^2}{2},$$

that is, $p_1 - p_2 = w$ and $p_1 p_2 = \frac{1}{2}(v^2 - w^2)$.

The distance is d each way, so the outward time is $t = \dfrac{d}{\sqrt{2}\,p_1}$ and the correct return time is $\dfrac{d}{\sqrt{2}\,p_2}$. Hence

$$\begin{aligned}\frac{d}{\sqrt{2}\,p_2} &= t + \frac{d}{\sqrt{2}}\left(\frac{1}{p_2} - \frac{1}{p_1}\right) = t + \frac{d}{\sqrt{2}} \cdot \frac{p_1 - p_2}{p_1 p_2} \\ &= t + \frac{d}{\sqrt{2}} \cdot \frac{2w}{v^2 - w^2} = t + \frac{\sqrt{2}\,wd}{v^2 - w^2} \text{ hours,}\end{aligned}$$

which is the required result.

Checks

- If $w = 0$ both return times reduce to t, as they should in still air.
- With $v = 300$, $w = 60$, $d = 400$: $t \approx 1.1785\,\text{h}$, the wrong return (found by stepping the motion numerically) takes $1.7678\,\text{h} = \frac{360}{240}t$, and the correct return takes $1.5713\,\text{h}$, matching $t + \frac{\sqrt{2}\cdot 60 \cdot 400}{86400}$. Random trials over many $v > w$ and d agree to rounding error.
- The wrong return is always slower than the correct one: this is the inequality $\frac{2wt}{v-w} > \frac{\sqrt{2}\,wd}{v^2 - w^2}$, i.e. $\sqrt{2}(v + w)t > d$, which holds because the outward ground speed $\frac{d}{t} = \sqrt{2}p_1 = \sqrt{2}(p_2 + w)$ is less than $\sqrt{2}(v + w)$, as $p_2 < v$.

Common slips

- Reversing the ground velocity instead of the air velocity: the wind does not turn round when the pilot does.
- Taking the wind as blowing *towards* the West. "From the West" means it carries the plane east.
- Solving each quadratic with the formula and subtracting the reciprocals of messy surds; spotting that p_1 and $-p_2$ are roots of one quadratic avoids all of that.

Beyond STEP

This is relative velocity in its simplest form: the velocity of the plane in the ground frame is its velocity in the air's frame plus the velocity of the air, a Galilean change of frame. The same bookkeeping, done carefully, underlies navigation, the river-crossing problems of first-year mechanics, and (when it breaks down at high speed) the motivation for special relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats moving frames thoroughly.

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